

The Components of the Soul

How Genetics, Epigenetics, Lifelong Learning, Mind-at-Large, and the Dissociated Mind Form an Individual Soul

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Date: September 1, 2026

Document Status: 🟡 [WORKING DRAFT • WORK IN PROGRESS]

Theoretical Architecture: The Conative-Integrative Framework (CIF)

Working Paper / Work in Progress: Working Paper / Work in Progress:

This treatise represents an active working draft and exploratory conceptual model within *The Conative-Integrative Framework (CIF)*. It is subject to ongoing research, empirical formalization, and philosophical refinement.

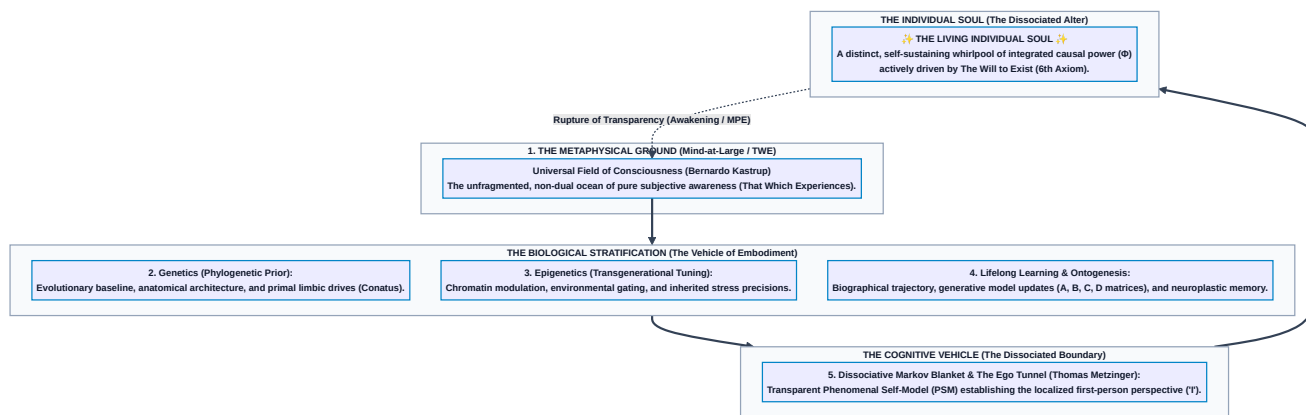


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Executive Summary: De-Mystifying the Soul

Throughout human history, the concept of the “**Soul**” (*Seele* / *Anima* / *Psyche*) has oscillated between two unsatisfactory extremes: 1. **Supernatural Dualism**: Treating the soul as an immaterial, magical “ghost” awkwardly inserted into a mechanical meat-robot. 2. **Eliminative Materialism**: Reducing the soul to a complete illusion—an accidental byproduct of dead matter with no genuine depth, purpose, or ontological reality.

Within **The Conative-Integrative Framework (CIF)**, we formulate a rigorous, post-materialist and naturalistic definition:

The Individual Soul is a localized, self-sustaining whirlpool (*dissociated alter*) within the universal field of Mind-at-Large. It is uniquely shaped by the fivefold intersection of: Universal Consciousness, Phylogenetic Genetics, Transgenerational Epigenetics, Biographical Lifelong Learning, and the Dissociative Boundary of the Phenomenal Self-Model (PSM).

The soul is neither magical nor an illusion; it is **the concrete 1st-person interiority of an autopoietic living network actively sustaining its integrated causal power (Φ) across time.**

1. The Ontological Ground: Mind-at-Large (TWE)

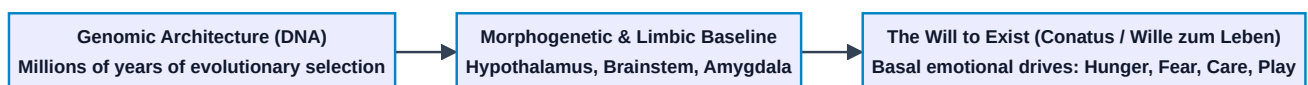
In Bernardo Kastrup's **Analytic Idealism**, the ultimate reality of nature is mental: there is only one universal field of consciousness—**Mind-at-Large** or **That Which Experiences (TWE)**.

- The cosmos is not made of tiny inanimate building blocks that mysteriously generate consciousness.
- Rather, **all physical matter is what mental processes look like when observed from across a dissociative boundary.**
- Mind-at-Large provides the **boundless subjective potentiality** (the pure water of the ocean).

Without Mind-at-Large, there would be no *subjectivity* anywhere in the cosmos. Mind-at-Large is the shared, irreducible qualitative substrate of every living entity.

2. The Phylogenetic Anchor: Genetics as the Ancient Prior

How does universal consciousness take on a specific biological form? Through the **Phylogenetic Prior**:



1. Evolutionary Priors (D_0 in Active Inference):

The genetic sequence is not merely a recipe for proteins; in predictive processing, it establishes the **deep evolutionary generative priors** of the organism. It encodes the expected homeostatic ranges (temperature, blood sugar, oxygen) necessary for physical survival.

2. The Limbic Seat of Will (Gerhard Roth):

Genetics hardwires the deep subcortical structures (brainstem, hypothalamus, amygdala) that generate what Spinoza termed **Conatus**, Schopenhauer termed the **Wille zum Leben**, and our framework defines as **The Will to Exist**.

3. Temperament Baseline:

Genetics provides the initial affective temperament (introversion/extraversion, novelty seeking, stress threshold) that forms the baseline coloration of the individual soul.

3. The Transgenerational Resonance: Epigenetics as Environmental

Gating

Genetics alone is too rigid to capture the adaptive dynamism of life. The bridge between ancestral history and immediate reality is **Epigenetics**:

- **Molecular Memory (DNA Methylation & Histone Modification):**

Epigenetics regulates gene expression without altering the underlying genetic code. It acts as a dynamic volume dial on genetic potential.

- **Transgenerational Trauma and Resilience:**

Environmental pressures, existential shocks, nutritional deprivations, and profound emotional experiences endured by parents and grandparents leave chemical imprints on the germline and developmental epigenome.

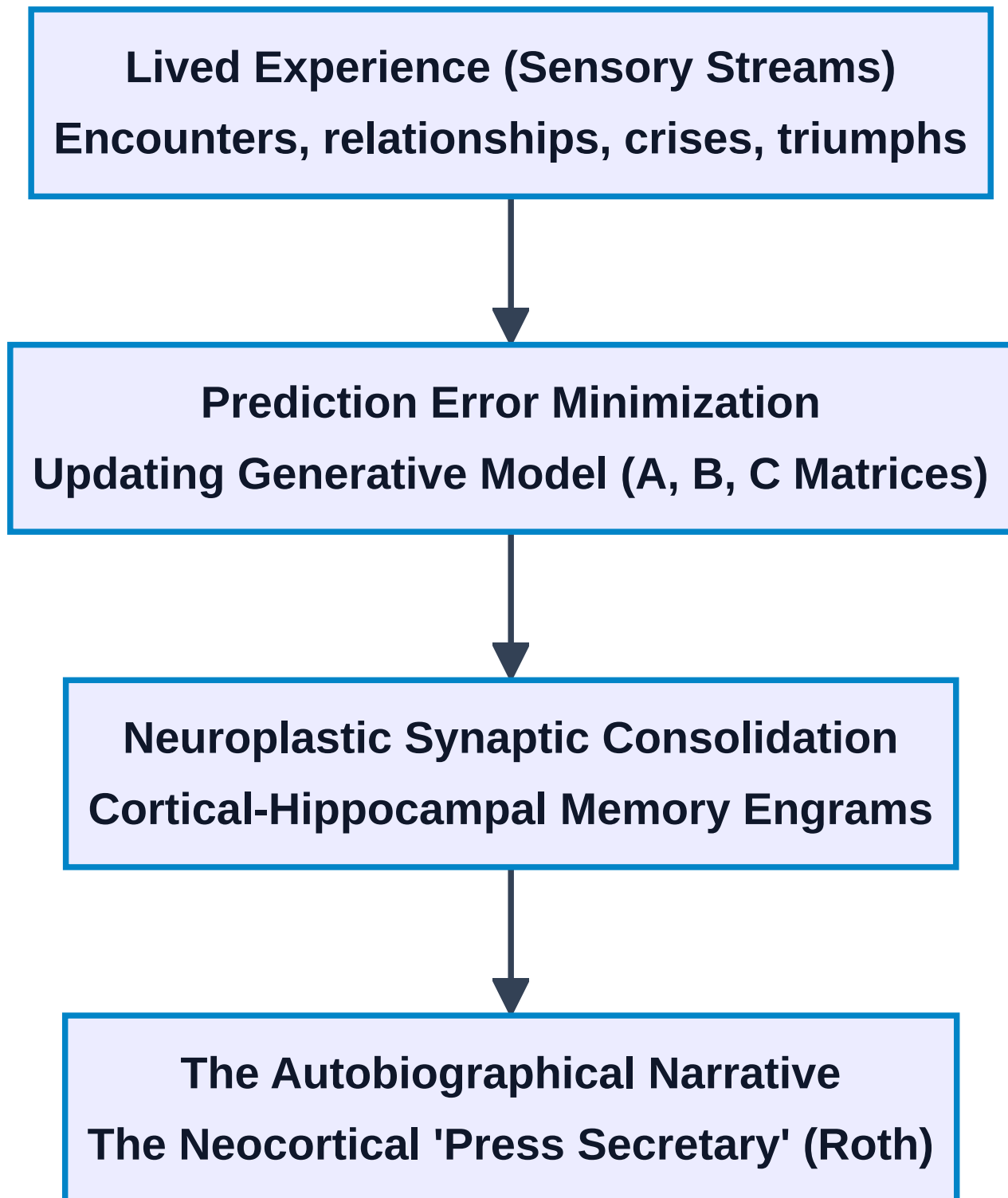
- **Predictive Precision Tuning:**

In Active Inference terms, epigenetics adjusts the **prior precisions** (γ) of the sensory apparatus. An organism born from a lineage that survived famine or danger inherits heightened vigilance and hyper-sensitive stress responses.

Epigenetics explains why the individual soul enters the world with **pre-configured intuitions, deep-seated sensitivities, and somatic inclinations** that precede its own individual biography.

4. The Ontogenetic Tapestry: Lifelong Learning and the Predictive Self

While genetics and epigenetics provide the canvas and initial palette, the **unique biographical portrait of the soul is painted through lifelong learning and ontogenesis**:



1. Continuous Updating of the Generative Model:

Every interaction with the world updates the agent's internal Transition Tensor (B) and Likelihood Matrix (A). The soul learns what causes pain, what brings joy, and what sustains coherence.

2. Memory Engrams and Biographical History:

Personal trauma, formative relationships, intellectual discoveries, and cultural immersions are

physically etched into synaptic weight distributions and cortical networks.

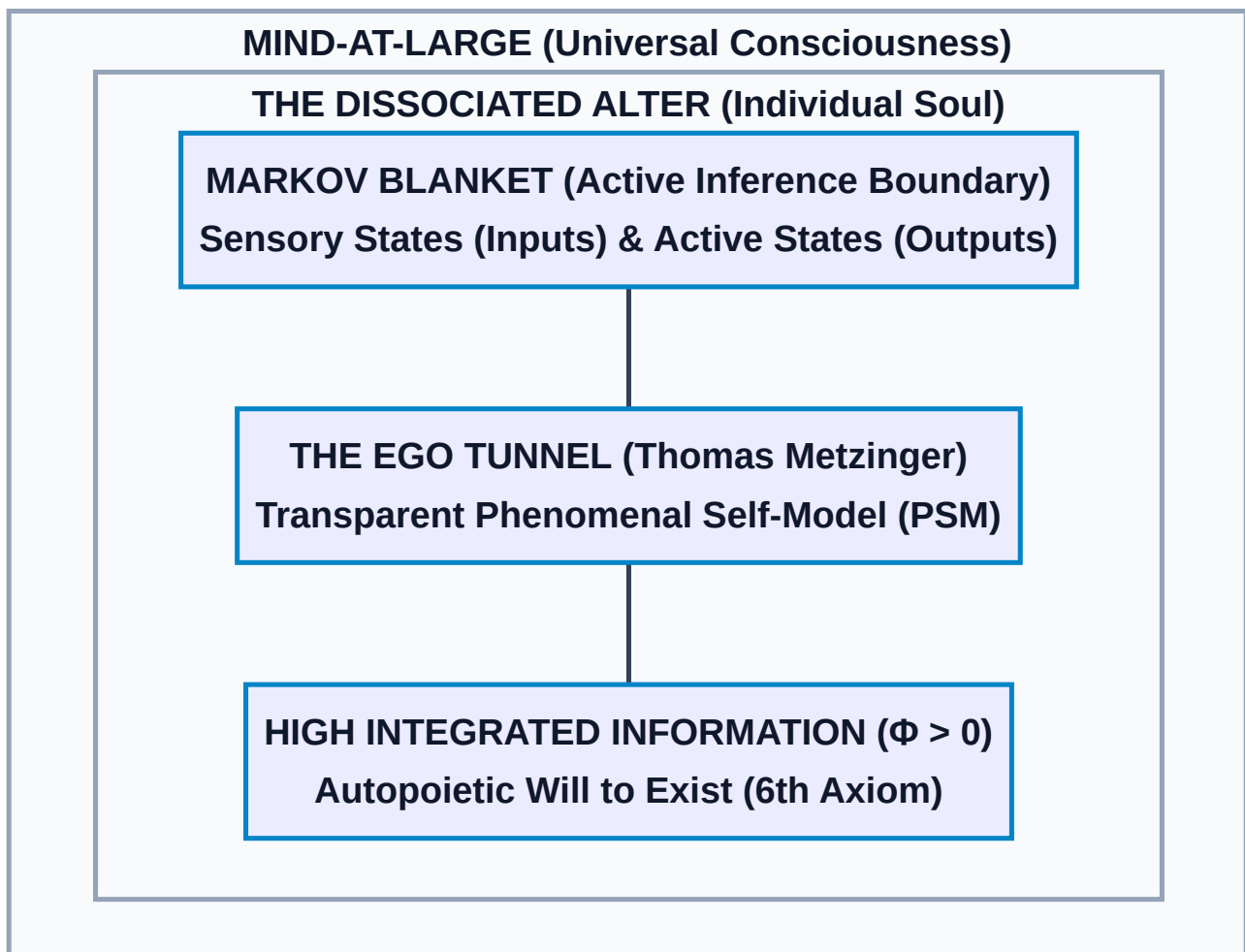
3. The Narrative Self (Gerhard Roth's "Press Secretary"):

The neocortex weaves these distributed memories into a continuous, coherent story: „*I am someone who survived bankruptcy; I am someone who raised children in Cameroon; I am someone who seeks the ultimate truth of mind.*“

This layer gives the individual soul its **unrepeatable individuality, wisdom, character, and psychological depth**.

5. The Dissociative Boundary: Markov Blankets and the Ego Tunnel

Why does the individual soul feel separate from the rest of the universe?



1. The Dissoziative Markov Blanket (Karl Friston):

A living organism maintains a boundary of internal states separated from external states by

sensory and active states. Statistically, this boundary creates a **phase-space partition** at the *Edge of Chaos*.

2. **The Transparent Phenomenal Self-Model (Thomas Metzinger):**

The brain generates a simulation of the organism as a whole within its world-model. Because the brain cannot introspect the fast, sub-symbolic computation generating this model, the model is **transparent**: the agent does not experience a model of a self, but experiences *being a self* inside an **Ego Tunnel**.

3. **The Phenomenological Illusion of Dualism:**

The transparent PSM is the cognitive mechanism that causes Mind-at-Large to experience itself as a localized, vulnerable, striving individual ego (*Alter*).

6. The Synthesis: The Architecture of the Individual Soul

Combining all five components yields the **Structural Equation of the Individual Soul**:

Individual Soul =

Mind-at-Large

Ontological Ground (TWE)

∩

Genetics

Phylogenetic Prior

∩

Epigenetics

TransgenerationalMemory

∩

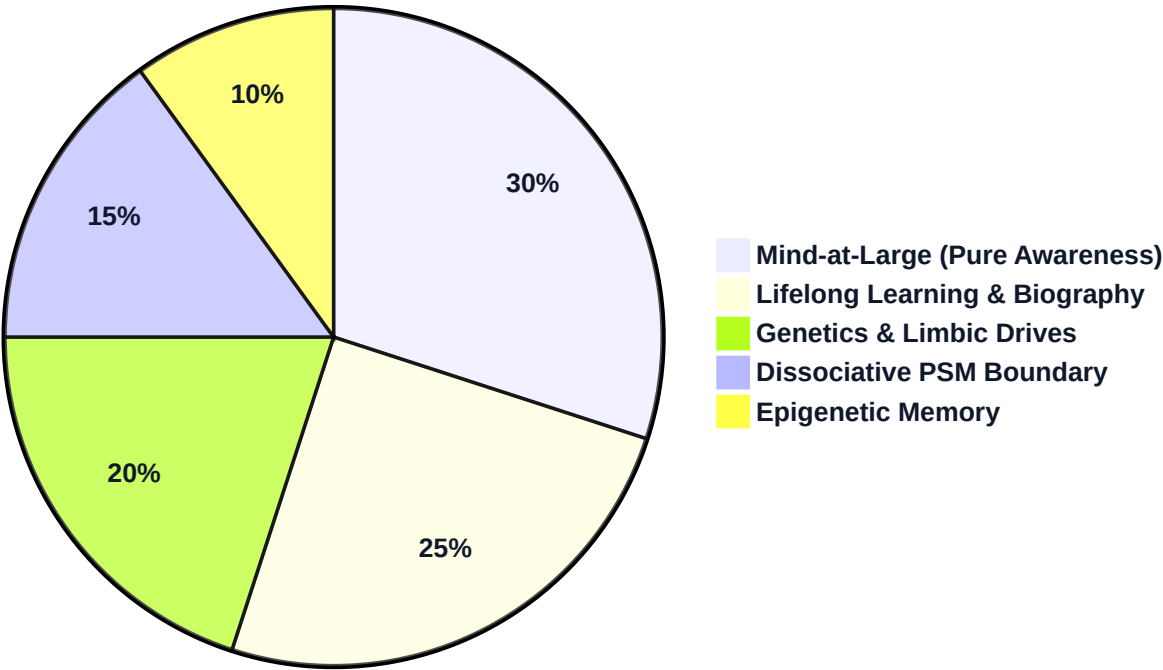
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The 5 Components of the Soul at a Glance:

Component	Scientific / Philosophical Domain	Role in the Formation of the Soul	Formal Framework Metric
1. Mind-at-Large	Analytic Idealism (Kastrup)	The irreducible qualitative ground; pure capacity for experience	That Which Experiences (TWE)
2. Genetics	Evolutionary Biology (Spinoza / Schopenhauer)	Physical blueprint, basal temperament, and primal <i>Will to Exist</i>	Baseline Priors (D_0) & Limbic Conatus
3. Epigenetics	Molecular Epigenetics & Epigenomics	Transgenerational tuning, ancestral vigilance, and stress thresholds	Precision Weighting Parameters (γ)
4. Learning	Cognitive Neuroscience (Roth)	Biographical memory, personal history, and generative model updates	POMDP Matrices (A, B, C) & Synaptic Engrams
5. Dissociation	Self-Model Theory & FEP (Metzinger / Friston)	The Markov Blanket & transparent PSM creating the localized "I"	Integrated Causal Power (Φ) across Boundary

Composition of the Individual Soul's Character



7. Death, Dissolution, and Minimal Phenomenal Experience (MPE)

What happens when the physical vehicle dies, or when consciousness undergoes profound mystical/non-dual realization?

1. Mystical Awakening / Minimal Phenomenal Experience (MPE):

When meditative inquiry, psychedelics, or deep philosophical insight destabilize the **transparency of the Ego Tunnel**, the system experiences what Thomas Metzinger calls **Minimal Phenomenal Experience (MPE)**: * The cognitive boundary (PSM) becomes **opaque**—recognized as merely a predictive model. * The illusion of the separate ego dissolves. * The individual soul recognizes its ultimate identity: „*I am not merely the transient alter inside the tunnel; I am the underlying field of Mind-at-Large experiencing itself.*“

2. Biological Death (Physical Dissolution):

When the biological organism dies, the Markov Blanket breaks down and the internal cause-effect structure collapses ($\Phi \rightarrow 0$). * **What dies:** The localized dissociative boundary (the PSM), the genetic machinery, and the active predictive loop. * **What remains:** Just as a whirlpool slows down and merges back into the ocean, the localized mental activity dissolves back into the infinite, unbroken field of **Mind-at-Large**. * The qualitative experiences, learning, and systemic evolution are not “lost” into voidness, but represent the historical crystallization through which Mind-at-Large has experienced and known itself.

8. References & Academic Bibliography

1. **Friston, K. (2010).** *The free-energy principle: a unified brain theory?* Nature Reviews Neuroscience, 11(2), 127–138.
2. **Jablonka, E., & Lamb, M. J. (2014).** *Evolution in Four Dimensions: Genetic, Epigenetic, Behavioral, and Symbolic Variation in the History of Life.* MIT Press.
3. **Kastrup, B. (2019).** *The Idea of the World: A Multi-Disciplinary Argument for the Mental Nature of Reality.* Iff Books.
4. **Kastrup, B. (2021).** *Science Ideated: The Fall of Matter and the Contours of the Next Mainstream Scientific Worldview.* Iff Books.
5. **Metzinger, T. (2003).** *Being No One: The Self-Model Theory of Subjectivity.* MIT Press.
6. **Metzinger, T. (2009).** *The Ego Tunnel: The Science of the Mind and the Myth of the Self.* Basic Books.
7. **Metzinger, T. (2024).** *The Elephant and the Blind: The Experience of Pure Consciousness: Philosophy, Science, and 500+ Experiential Reports.* MIT Press.

8. **Riebl, T. (2026).** *The Conative-Integrative Framework (CIF): How Active Inference Networks, Integrated Information (Φ), and the 6th Axiom Fit Together to Unite Analytic Idealism, the Free Energy Principle, and Consciousness.* Master Paper & Working Treatise, Luxembourg.

9. **Roth, G. (2003).** *Aus Sicht des Gehirns.* Suhrkamp Verlag, Frankfurt am Main.

10. **Roth, G. (2021).** *Wie das Gehirn die Seele macht: Emotionen, Bewusstsein, Unbewusstes.* Klett-Cotta, Stuttgart.

11. **Schopenhauer, A. (1819/1844).** *Die Welt als Wille und Vorstellung (The World as Will and Representation).* F. A. Brockhaus, Leipzig.

12. **Spinoza, B. (1677).** *Ethica, ordine geometrico demonstrata.* Posthumous Publication.

13. **Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016).** *Integrated information theory: from consciousness to its physical substrate.* Nature Reviews Neuroscience, 17(7), 450–461.

14. **Yehuda, R., & Lehrner, A. (2018).** *Intergenerational transmission of trauma effects: putative role of epigenetic mechanisms.* World Psychiatry, 17(3), 243–257.

9. Tool Attribution & Colophon

Tooling Colophon: Tooling Colophon:

This theoretical treatise, philosophical architecture, and scientific synthesis were conceptualized and authored by **Thomas Riebl** (Luxembourg) as part of **The Conative-Integrative Framework (CIF)**.

The conceptual formulation, structural structuring, vector diagram styling, and multi-format document compilation (Word .docx , Print-Ready PDF, and Markdown) were developed with the assistance of **Google Gemini (Antigravity Advanced Agentic Coding System)** (September 2026).

Vault & Repository References

- **Master Framework Paper (CIF):** /home/thr/Documents/The_Conative_Integrative_Framework_Thomas_Riebl.pdf
- **GitHub Project Repository:** <https://github.com/Thrieb1/active-inference-phi-network>
- **Vault Archive:** </home/thr/Documents/ThRNotes/03-professional/braindumps/>